MILITARY SPECIFICATION

TEST SET, ELECTRONIC CIRCUIT PLUG-IN UNIT: AN/GVM-8

1. SCOPE

1.1 This specification covers an electronic test set which is used for detecting defective modules of the Rangefinder Laser, AN/GVS-3 (see 6.1) after they have been removed from the rangefinder.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-I-46058	Insulating Compound, Electrical, for Coating Printed Circuit Assemblies
MIL-P-46843	Printed Circuit Assemblies, Design and Production of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U. S. Army, Frankford Arsenal

A10555677	Test Set, Electronic Circuit Plug-In Unit AN/GVM-8

PACKAGING DATA SHEET

10555677	Packaging of Test Set, Electronic Circuit Plug-In Unit AN/GVM-8

FSC 4931

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(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with the drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication.- The Test Set, Electronic Circuit Plug-In Unit, AN/GVM-8, hereinafter referred to as test set, shall be manufactured in accordance with Drawing A10555677 and drawings pertaining thereto.

3.3 General specification.- The test set shall meet the following requirements, where applicable, of MIL-F-13926 and MIL-P-46843:

a. Dimensions and tolerances.
b. Inorganic protection surface finisher.
c. Electrical and electronic assemblies.
d. Soldering electrical connections.
e. Part identification and marking.
f. Workmanship.

3.3.1 Electronic workmanship.- In addition to meeting the workmanship requirements of MIL-F-13926 and MIL-P-46843, the test set shall meet the electronic workmanship requirements of 3.3.1.1 through 3.3.1.11.4, inclusive.

3.3.1.1 Circuit board workmanship.

3.3.1.1.1 Spacing of parts.- Unless otherwise specified on the applicable drawing, all parts shall be mounted with at least one point of the component in contact with the board. The component shall not be raised more than 0.031 inch from the board at any point on its contacting surface.

3.3.1.1.2 Symbols and markings.- Unless physically impossible, all parts shall be mounted on the board in such a manner that the proper identification markings on each part are visible.

3.3.1.1.3 Transistors.- Transistors are to be mounted perpendicular to the circuit board with spacers between the transistors and the circuit board.

3.3.1.1.4 Bend radius.- The minimum bend radii for formed leads shall be one lead diameter. The maximum shall be two diameters or 0.062 inch, whichever is smaller.

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3.3.1.1.5 Component leads.- The minimum lead length for radial lead components shall be 4 times the lead diameter. The portion of the lead between a bend and the component body shall emerge from the body in a straight line radially from the component body and enter the plated hole approximately perpendicular to the board. Unless otherwise specified in the applicable drawing, leads of transistors mounted with a spacer directly into the printed circuit board, shall have a .062 inch of lead from the transistor base to any soldered joint. Component leads shall be preformed to suit mounting holes and clinched parallel to board after being inserted through the holes. There shall be visible evidence of leads passing completely through the circuit board. The leads shall not protrude more than .060 inch, including solder, beyond the side of the circuit board opposite the component. The leads shall not extend more than two lead diameters beyond the land area.

3.3.1.1.6 Soldering.- All solder joints shall conform to MIL-F-13926 and MIL-P-46843. There shall be a solder fillet between the board and the lead on the side of the board opposite the component. The fillet shall not obscure the contour of the clinched lead. A solder fillet is acceptable, but not required, on the component side of the board.

3.3.1.1.7 Plated-through connections.- All plated-through connections shall be filled with solder in order to eliminate dependence on plating itself for circuit continuity.

3.3.1.1.8 Electrical clearance.- A minimum of .010 inch shall be maintained between uncommon component leads, transistor cases and tabs.

3.3.1.1.9 Cleaning.- After soldering operations have been completed, the assembly shall be thoroughly cleaned and rinsed. The cleaning fluid shall have a gentle flow, at no time exceeding a pressure of 5 pounds per square inch (psi). The temperature of the cleaning solution shall not exceed 125 degrees Fahrenheit (°F). After cleaning there shall be no residual flux, surface contamination, or detrimental staining on the assembly. Cleaned assemblies shall not be allowed to become contaminated after cleaning, nor shall markings become illegible.

3.3.1.1.10 Drying.- The test set circuit board assemblies may be dried by exposure to room temperature without the aid of auxiliary heating or air circulating equipment. If other than the room temperature drying method is used, the assembly shall be dried in a circulating air oven, the temperature of which shall not exceed 165°F. When compressed air or nitrogen is used for drying parts, the pressure applied to the parts shall not exceed 5 psi. The air or nitrogen shall be filtered to meet the following requirements:

a. Permissible dust particle size not to exceed 6 microns.
b. Permissible moisture content limited to 2.12 grams per cubic foot (dew point 32°F maximum).
c. Dust count shall not exceed 1,000 particles per cubic foot.

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3.3.1.1.11 Circuit board coating compound.

3.3.1.1.11.1 Masking.- Prior to coating, the areas specified on the applicable drawings shall be masked. Subsequent to coating, with masking removed, all specified areas shall be clean and free of coating.

3.3.1.1.11.2 Coating.- The test set circuit boards shall be coated with insulating compound, electrical, type PUR polyurethane resin conforming to MIL-I-46058.

3.3.1.1.11.3 Thickness.- Multiple applications shall be used with proper curing between each coat, until a minimum of .0010 inch thick to .0060 maximum thickness is achieved.

3.3.1.1.11.4 Cured compound.- The cured compound shall present a smooth continuous surface and shall pass from the mounted components to the circuit board with no break at their junctions. The cured compound shall meet the following requirements:

a. Compound shall be free of visible dust particles or other foreign matter.
b. It shall be smooth and uniform and have no globules or bare spots.
c. It shall be substantially free of bubbles.
d. There shall be no blisters or cracks.
e. It shall be transparent and shall not affect marking legibility.
f. It shall be capable of being removed in local areas for replacement of components by means of soldering iron or similar method.
g. Applied to a test board a 1/16 inch thickness, the compound shall not rupture or separate from the board when the board is bent into an arc of twelve inches radius.

3.4 Environmental.

3.4.1 Vibration.- The test set shall withstand vibration for 15 minutes each in 3 mutually perpendicular directions. Vibration in each direction shall be at frequencies varied logarithmically from 5 to 500 cycles per second (cps). The double amplitude of applied vibration shall be 0.036 inch at lower frequencies and shall be maintained at, but not exceed, a 5g level at higher frequency. Subsequent to vibration, the test set shall show no evidence of physical damage, and shall meet the requirements of 3.4.5 through 3.6.

3.4.2 Shock.- The test set shall withstand a total of 18 shock impulses, 3 in each direction of 3 mutually perpendicular directions. Each shock impulse shall be a half-sine wave with a time duration of 11 ± 1 milli-seconds. The peak amplitude of each shock impulse shall be 15g's. Subsequent to shock, the test set shall show no evidence of physical damage and shall meet the requirements of 3.4.5 through 3.6.

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3.4.3 Storage temperatures.- The test set shall withstand thermal stabilization for a period of 5 hours at temperatures of -80°Fahrenheit (F) and +160°F. Subsequent to exposure to the temperatures specified herein, the test set shall show no evidence of physical damage, and shall meet the requirements of 3.4.5 through 3.6.

3.4.4 Immersion.- The test set with cover closed and locked shall show no evidence of leakage (gas bubbles from inside the unit) when the test set, which has been thermally stabilized at 75° ± 5°F, is completely immersed for a minimum of 12 minutes in water of 135° ± 5°F temperature and observed for the entire time period. Subsequent to immersion, the test set shall show no evidence of water penetration inside the unit and shall meet the requirements of 3.4.5 through 3.6.

3.4.5 Operational temperatures.- The test set shall meet the requirements of 3.5 through 3.6 when thermally stabilized at temperatures from -25°F to + 130°F inclusive.

3.5 Performance.- The test set shall be capable of meeting the requirements of 3.5.1 through 3.5.11 when energized with the following supply voltages:

110V, 60 or 400 Hz
28VDC

3.5.1 Power ON.- When the POWER switch is set to the ON position, the power ON indicator lamp shall be lighted and shall be adjustable for brightness. Subsequent depression of the PUSH TO TEST button shall allow Indicator Lamps A, B and C, which have built-in test functions, to be lighted and adjustable for brightness. Unless otherwise specified, power shall be supplied to any of the test set plug-in receptacles only while the PUSH TO TEST switch is depressed.

3.5.2 Counter tester.- When the TEST FUNCTIONS switch (S2) is set to the COUNTER position, depression of either the PUSH TO TEST switch to its first or second closure position shall cause the following results:

3.5.2.1 Counter assembly receptacle.

a. 9.0 ± 1.0 VDC shall be generated at Pin f and the meter shall deflect to a reading of 6 ± 1 divisions when a load of 15 ohms are connected between Pin f and Pin B.

b. 28 ± 2.0 VDC shall be generated at Pin e.

c. When the READ switch is set to the ON position and the high voltage interlock S5 located in the high voltage compartment is closed, DC levels of 16.5 ± 0.5V and 26.5 ± 0.5V shall be generated at Pins H and N respectively.

d. When the PUSH TO TEST switch is depressed to close its second set of contacts, the pulses shown in Figure 1 shall be generated at Pins Z, A, E and K. The amplitude of the pulse at Pin A shall be variable in amplitude from 0 volts to 4.0 ± 0.5 volts using the START adjustment of

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the RANGE ADJUSTMENT controls. The amplitude of the pulses generated at Pin E shall be variable in amplitude from 0 volts to 5.0 ± 0.5 volts using the STOP adjustment of the range adjustment control. The number of pulses generated at Pin E shall be variable from 0 to 6. Those pulses generated shall be selectable by using the toggle switches 1 through 6 of the RANGE ADJUSTMENT switches. The delay(s) between the pulse generated at Pin A and those generated at Pin E shall be variable for each pulse at Pin E, and shall be as specified below:

S_1 variable from 1.25 to 2.50 microseconds
S_2 variable from 2.40 to 6.50 microseconds
S_3 variable from 6.40 to 14.25 microseconds
S_4 variable from 11.00 to 29.00 microseconds
S_5 variable from 20.00 to 58.00 microseconds
S_6 variable from 50 to 140 microseconds

The variable delay for each pulse shall be controlled by the adjustment potentiometer adjacent to the toggle switch which selects the pulse to be generated.

e. Momentary application of a 3.0 ± 0.3 VDC level through a 1K ohm resistor to Pin R shall cause Indicator Lamp A to light.

f. Momentary application of a 3.0 ± 0.3 VDC level through a 1K ohm resistor to Pin X shall cause Indicator Lamp B to light.

g. Momentary connection of a ground potential to Pin W shall cause Indicator Lamp C to light.

h. Pins B, F, J and h shall be connected to chassis ground.

i. Pin C shall be connected to "Counter Assembly" TP 1.

j. Pin D shall be connected to "Counter Assembly" TP 2.

k. Pin S shall be connected to "Counter Assembly" TP 3.

l. Pin T shall be connected to "Counter Assembly" TP 4.

m. Pin L shall be connected to chassis ground through a 10 meg ohm ± 10% resistance.

3.5.2.2 Connector 10547210.

a. 5.5 ± 0.3 VDC shall be generated at Pin c when Switch IX is set to position A.

b. No output voltage shall be present at Pin c when Switch IX is in position B.

c. 3.75 ± 0.25 VDC shall be generated to Pin W.

d. 4.0 ± 0.5 VDC shall be generated to Pin Y and shall be removed only while Switch II is depressed.

e. 4.0 ± 0.5 VDC shall be generated to Pin D and shall be removed only while Switch II is depressed.

f. The voltages above shall reappear at Pins Y and D when Switch II is released. As Switch II is released a positive pulse with characteristics described in Figure 2a shall appear at Pin o across a 1K ohm load.

g. 4.0V ± 0.5 shall be generated through a 3.3K ± 5% resistance to Pin n when Switch VIII is set to position A. Pin n shall become connected

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to chassis ground when Switch VIII is set to position B. Pin F shall have no output when Switch VI is in the AUTO position. When switch is set to MAN position and Switch III is depressed a negative pulse having the characteristics shown in Figure 2b shall be generated at Pin F.

h. When the READ switch is set to the ON position, connecting Pins Q, G, O, E, M, C, A, N, B or P to chassis ground shall cause the numeral 0, 1, 2, 3, 4, 5, 6, 7, 8 or 9 respectively to light on the DIGIT INDICATOR.

i. Pin H shall be connected to chassis ground through a 1K ± 5% ohm resistance.

j. Pin J shall be connected to chassis ground through a 4.7K ± 5% ohm resistance.

k. Pin L shall be connected to chassis ground through a 4.7K ± 5% ohm resistance.

l. Pin 1 shall be connected directly to chassis ground.

3.5.2.3 Connector 10547220.

a. 3.75 ± 0.25VDC shall be generated at Pin c.

b. 4.0 ± 0.5VDC shall be generated at Pin Q. When Switch II is depressed, this DC output shall not be present.

c. With Switch IV in position A, a 1.8 ± 0.15VDC level shall be generated to a 1K ± 5% ohm load at Pin W.

d. With Switch IV in position B, a 1.8 ± 0.15VDC level shall be generated to a 1K ± 5% ohm load at Pin U.

e. With Switch IV in the center position there shall be no output at either Pin U or Pin W.

f. With Switch VIII set to position A, 3.75 ± 0.25VDC shall be supplied to Pin N through a 3.3K ± 10% resistor.

g. With Switch VIII in position B, Pin N shall be connected to chassis ground.

h. As Switch I is depressed, a pulse as described in Figure 3a shall be generated to a 1K ohm load at Pin K, and a 3.75 ± 0.25VDC level shall be generated at Pin Z and Pin G.

i. When Switch II is depressed there shall be no output at Pin H. As Switch II is released, a pulse as specified in Figure 3b shall appear at Pin H.

j. With Switch VI in the manual (MAN) position, 3.75 ± 0.25VDC shall be generated through a 1.3K ohm resistor to Pin J. As Switch III is depressed, a negative pulse as described in Figure 3c shall be generated at Pin J across a 1K load.

k. With Switch VI set in the AUTO position, a train of pulses as described in Figure 3d shall be generated at Pin J across a 1K load.

l. With the READ switch set to the ON position, the numeral 0, 1, 2, 3, 4, 5, 6, 7, 8 or 9 of the DIGIT INDICATOR shall be capable of being lighted as Pins n, Y o, X, m, V, k, T, i or S respectively is connected to ground.

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m. The following resistive loads shall exist between the designated pin and chassis ground:

Connector Pin	Load	Connector Pin	Load
Pin A	1K	Pin b	1K
L	2K	I	1K
B	10K	E	1K
D	2K	C	2K
0	1.5K	P	1.5K
f	2K		
d	1K		

n. Pin F shall be directly connected to TP21.
o. Pin 1 shall be connected to chassis ground.

3.5.2.4 Connector 10547230.

a. 28 ± 2.0VDC shall be generated to Pin O.
b. 9 ± 1,0VDC shall be generated to Pin o.
c. A pulse as specified in Figure 4a shall be generated at Pin J when Switch IV is set to position A across a 1K load.
d. A pulse as specified in Figure 4b shall be generated at Pin I when Switch IV is set to position B across a 1K load.
e. A 3.75 ± 0.25 VDC level shall be supplied to Pin K when Switch VIII is in position A.
f. A 1K ± 5% resistance shall be connected from Pin K to chassis ground when Switch VIII is set to position B.
g. When Switch VI is set to the AUTO position a square wave output as shown in Figure 4d shall be generated at Pin E.
h. When Switch VI is set to the MAN position, a single pulse as shown in Figure 4c shall be generated at Pin E each time Switch III is depressed.
i. When Switch I is depressed the DC voltage output at Pin O shall drop to 16 ± 2.0.
j. With the READ switch in the ON position, alternately Connecting Pins b, a, S, W. R, V, Q, Z, P or T to chassis ground shall cause the numeral 0, 1, 2, 3, 4, 5, 6, 7, 8 or 9 respectively of the DIGIT INDICATOR to be lighted. Pin U shall be connected to the internal 160 ± 5VDC source through a 68K ± 5% resistance.
k. The following resistances shall be exist from the indicated pin to chassis ground.

Connector Pin	Resistance to Ground (ohms)
G	12
Y	24
X	33
N	120
B	3K
L	75
D	3K
F	1K
C	2K
k	0

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3.5.2.5 Connector 10547240.

a. 3.5 ± 0.3VDC shall be generated at Pin k.
b. 3.75 ± 0.25VDC shall be generated at Pin G.
c. 3.75 ± 0.25 VDC shall be connected to Pin A through a 4.7K ± 5% resistance.
d. 3.75 ± 0.25VDC shall be connected to Pin K through a 1.0K ± 5% resistance.
e. When Switch II is depressed the 3.5VDC output shall be removed from Pin k.
f. When Switch II is depressed and released, a pulse as specified in Figure 5a shall be generated at Pin B.
g. When Switch I is depressed and released, a pulse as specified in Figure 5b shall be generated to a 1K ohm ± 5% load at Pin b.
h. As Switch IV is set to position B, a pulse as specified in Figure 5c shall be generated to a 1K ohm ± 5% load at Pin V.
i. As Switch IV is set to position A, a pulse as specified in Figure 5d shall be generated to a 1K ohm ± 5% load at Pin R.
j. When Switch V is set to position P, Q, W or Y, 3.75 ± .25VDC shall be connected through a 3K ohm ± 5% resistance to Pin P, Q, W or Y respectively on the connector.
k. When Switch VII is set to position A, Pin E shall be connected to chassis ground.
l. When Switch VII is set to position B, P E shall have an open circuit to chassis ground.
m. When Switch VIII is set to position A, Pin O shall be connected to 3.75 ± .25 VDC through a 3.3K ohm ± 5% resistance.
n. When Switch VIII is set to position B, Pin O shall be connected to chassis ground.
o. When Switch VI is set to man position, Pin X shall be connected to 3.75 ± .25VDC source through a 24K ohm resistor.
p. As Switch III is depressed, a negative pulse as specified in Figure 5e shall be generated at Pin c across a 1K load.
q. Pin L shall be connected thru a 470 ohm resistor to the base of a NPN transistor which is connected to ground in common emitter configuration.
r. The voltage at Pin K shall drop from a d.c. level of 3.75 ± .25 volts to a d.c. level of 0.3 ± 0.1 volts when a voltage of 2.0 ± 0.5 volts is applied to Pin L.
s. With the READ switch set to the ON position, the digits 0, 1, 5 or 7 of the DIGIT INDICATOR shall be lighted when Pin Z, a, h, or U respectively is connected to chassis ground.
t. Neon indicator lamp DS1, DS2, DS3, DS4 or DS5 shall be individually lighted when Pin I, H, F, J or d respectively is connected to chassis ground.
u. Pin M shall be connected to ground through a 4.7K ± 10% resistance.
v. Pin i shall be connected to ground through a 1K ± 10% resistance.
w. Pin o shall be directly connected to chassis ground.

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x. The following connections shall exist between the pins and test points as indicated below:

Connector Pin	Test Point
C	17
e	21
j	22
T	23
N	24
D	27

y. When the lever switch is depressed, Indicator Lamps A, B and C shall be capable of being lighted using their built-in test functions and shall be adjustable for brightness.

3.5.3 Stop Pre Amp and PM Tube Tests.- When the TEST FUNCTIONS switch (S2) is set to the STOP PRE AMP and PM TUBE position, the following voltages and signals shall be generated by the test set.

a. Output at Jack J10. Pin A 5.5 ± 0.3V capable of supplying 50 ma.

b. Output at Jack J10 Pin C across a 180 ohm load shall be as shown in Figure 6a.

c. A group of two negative going pulses each having the charateristics defined in Figure 6b shall be generated at Connector J14. These pulses shall be adjustable in peak amplitude from .01V to .6V by the AMPLITUDE adjustment potentiometer and shall be adjustable in separation from 8 ± .5 microseconds to 2 ± .5 microseconds by the SEPARATION adjustment potentiometer. These groups of pulses shall occur at a repetition rate of 800 ± 200 times per second. Each pulse shall have a pulse width of 125 ± 25 nanoseconds at the half amplitude point.

d. Light output.- The light emitting diode (LED) located in the PM tube receptacle shall generate pulses of light occurring in groups of two with a group repetition rate of 800 ± 200 hz. The spacing between pulses shall be variable using the SEPARATION adjustment potentiometer. Each pulse of light shall have a pulse width of 50 ± 20 nanoseconds at the half amplitude point and emit 10 ± 2 milliwatts of power in a spectral bandwidth of 400 Angstroms (A) centered around 6800A.

e. High voltage output.- (Negative Polarity) A voltage of 1400 ± 200 VDC shall be generated at the HIGH VOLTAGE connector when the interlocking switch in the PM TUBE receptacle is depressed. The center contact of the connector shall be the negative terminal.

f. A resistance of 330 ± 5% ohms shall exist from Pin E on Connector J10 to chassis ground, and electrical continuity shall exist from Pin E of Connector J10 to the center contact of the PRE-AMP OUTPUT connector.

g. A resistive load of 68 ± 5% ohms shall exist from the PM TUBE OUTPUT connector to chassis ground and electrical continuity shall exist from the center contact the PM TUBE OUTPUT connector to the center contact of the OUTPUT connector.

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3.5.4 Nixie power supply test.- When the TEST FUNCTIONS switch is set to the NIXIE POWER SUPPLY position, the following shall occur:

a. An output voltage of 28 ± 2.0VDC shall be generated between Pins A, X (+) and B (-) of the NIXIE POWER SUPPLY Receptacle (J4).

b. Application of 160 ± 1VDC between Pins E (+) and B (-) of J4 shall cause the panel meter of the test to deflect to 6.5 ± .5 Divisions.

c. A resistance of 22,000 ± 5% shall exist between Pins E and B of J4.

d. Electrical continuity shall exist between Pin B of J4 and chassis ground.

3.5.5 Pulse forming network power supply test.- When the TEST FUNCTION switch is set to the PFN POWER SUPPLY position, the following shall occur:

a. 5.5 ± 0.3VDC shall be supplied through a 5.1K ± 5% ohm resistance to Pin H of the PFN POWER SUPPLY receptacle.

b. When the PUSH TO TEST switch is depressed to close only the contacts of its first closure position, an output voltage of 28 ± 2.0VDC shall be generated at J7 between Pins A (+) and B (-), and between Pins C (+) and F (-).

c. Indicator Lamp A shall be capable of being lighted when a 5.5 ± 0.3VDC level is applied to Pin L through a 5.1K ohm resistance.

d. When the PUSH TO TEST switch is depressed to close its second set of contacts, the connector body of J5 shall be connected to Pin D through a parallel wired RC network with component values R = 100K ohms, C = 0.22 microfarads.

e. When the PUSH TO TEST switch held completely closed, the application of 1500 ± 1% VDC to the co-axial connector of the PFN power supply test position shall cause the meter to deflect 6 ± 1/2 divisions.

f. When the 1500VDC input is subsequently removed, the meter shall not fall below a reading of 5 divisions for at least 5 seconds after the 1500VDC input is removed. Subsequent release of the PUSH TO TEST switch shall allow the meter to return to a reading of 0 within 5 seconds after release.

g. A resistance of 16K ohms shall exist from Pin K of J7 to chassis ground. A resistance greater than 100K ohms shall exist from the outer conductor of J5 to ground.

3.5.6 Photo multiplier power supply.- When the TEST FUNCTION switch is set to the PM POWER SUPPLY position, the following shall occur:

a. When the PUSH TO TEST switch is depressed to close only the contacts of its first closure position, an output voltage of 28 ± 2.0VDC shall be established between Pins A (+) and B (-) of the PM POWER Supply Receptacle J9, and there shall be no output at Pins C or E of J9.

b. As the PUSH TO TEST switch is depressed to close the second set of contacts, a positive pulse as shown in Figure 7b shall be generated to a 1K ohm load at Pin C of J9. Subsequently, as the PUSH TO TEST switch is partially released to close only the 1st set of contacts, a negative pulse as shown in Figure 7a shall be generated at Pin E of J9.

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c. Application of 1500 ± 1% VDC to J8 with the negative side of this voltage applied to the center conductor and the positive side applied to the connector body shall cause the meter to deflect to 6 ± 1/2 divisions.

3.5.7 Q-Switch timing circuit.- When the TEST FUNCTION switch is set to the QSTC position, the following shall occur:

a. When the PUSH TO TEST switch is depressed to close only the contacts of its first closure position, an output voltage of 28 ± 2.0VDC shall be established between Pins E (+) and B (-) of the QSTC Receptacle J6.

b. When the PUSH TO TEST switch is depressed to close the second set of contacts, a positive pulse as described in Figure 8a shall be generated to a 1K ohm load at Pin A of J6.

c. With 20 ± 0.2VDC applied across Pins D (+) and C (-) the waveform shown in Figure 8b shall be generated across Pins J and L (GND). When a level of 3.0 ± 0.3VDC is momentarily applied to Pin F, Indicator Lamp b shall light.

d. When a level of 3.0 ± 0.3VDC is momentarily applied to the center conductor of J17, indicator lamp a shall light.

e. When a ground potential is momentarily applied to Pin H of J6, Lamp c shall light.

3.5.8 Cavity.- When the TEST FUNCTION switch is set to the CAVITY position, the following shall occur:

a. A DC voltage of 5.0 ± 0.5V shall be generated at Pin J of Receptacle J18 and a +5.5 ± 0.3VDC level shall be established at Pin F of J18.

b. When Pin K of J18 is connected to chassis ground, the meter shall deflect to a reading of 5 ± 1 divisions.

c. When the PUSH TO TEST switch is depressed to close only the contacts of its first closure position, an output voltage of 28 ± 2.0VDC shall be established at Pin D of J18.

d. When the PUSH TO TEST switch is depressed to close the second set of contacts, the 5.5V ± .03DC level shall be removed from Pin J of J18.

e. Pin A of J18 shall be connected to chassis ground.

3.5.9 Pulse forming network.- When the TEST FUNCTION switch is set to the PULSE FORMING NETWORK position, the following shall occur:

a. When the PUSH TO TEST switch is depressed to close only the contacts of its first closure position, an output voltage of 28 ± 2.0VDC shall be established at Pin E of J18.

b. When the PUSH TO TEST switch is depressed to open the second set of contacts, Pin L of J18 shall become disconnected from any source of voltage. Subsequently, when the PUSH TO TEST switch is partially released to again close the second set of switch contacts, a positive pulse as described in Figure 9 shall be generated to a 1K ohm at Pin L of J18.

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c. When a 250 microampere current is applied through Pin H of J18 to ground, the meter shall deflect to a reading of 5 ± 0.1 divisions.

3.5.10 Trigger module.- When the TEST FUNCTION switch is set to the TRIGGER MODULE position and the interlock switch of the high voltage compartment is closed, the following shall occur:

a. Depression of the PUSH TO TEST switch shall cause an output voltage of 160 ± 5VDC to be generated across the contacts of Terminal Block E1.

b. When the PUSH TO TEST switch is depressed to open the second set of contacts, Connector J3 shall become disconnected from any source of voltage. Subsequently, when the PUSH TO TEST switch is partially released to again close the second set of contacts, a positive pulse as described in Figure 10 shall be generated to a 1K ohm load connected to J3.

c. Application of 1000 ± 10VDC to Terminal P2 shall cause the Neon Bulb E to light.

3.5.11 High voltage divider.- When the TEST FUNCTION switch is set to the HIGH VOLTAGE DIVIDER position and the interlock switch of the high voltage compartment is closed, the following shall occur:

a. Depression of the PUSH TO TEST switch shall cause an output voltage of 1600 ± 100VDC to be generated at P3 with the center conductor the negative side of the output voltage. Connecting Terminal J16 to chassis ground shall cause Neon Indicator Lamp D to light.

b. With the PUSH TO TEST switch released and 160 ± 5VDC applied across E2 the meter shall deflect to a reading of 6.5 ± 0.5 divisions.

3.6 Accessory equipment.- The accessory items provided with the test set shall meet the requirements specified below:

3.6.1 PFN and Cavity Test Adapter, 11728010.

3.6.1.1 Cavity tests.- The test adapter shall perform as specified below when 5.0 ± 0.1VDC is applied across Pins F (+) and A (Gnd) of P5 and 28 ± 1VDC is applied between Pins D (+) and A (Gnd) of P5:

a. Indicator lamp.- The adapter shall compare the period of a square wave input applied to Pin B of J2 with an internal reference and Light DS2 when the period diminishes to 2.5 millisec ± 0.1 millisec.

b. Regulated output voltage.- 20 ± 2VDC shall be developed across a 4 ohm resistor connected between Pins D (+) and E (Gnd) of J2.

3.6.1.2 PFN test.- The test adapter shall perform as specified below when 28 ± 1VDC is applied across Pins M (+) and A (Gnd) of P5.

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a. Capacitor charging.- Each time S_1 is depressed and held, the test adapter shall charge a 75 microfarad ± 5% capacitor connected across the output of potted Transformer 10540823, to a voltage greater than 1400 volts but less than 1800 volts. The charge time shall not exceed 20 seconds.

b. Relay interlock.- When S_1 is depressed, the resistance measured across the contacts of P3 shall be greater than 10 megohms. When S_1 is not depressed, the resistance measured across P3 shall be 10K ohms ± 10%.

c. Flash lamp.- When a 75 microfarad ± 5% capacitor that is charged to 1500 ± 15VDC is connected across P3 (Pin A+) and S_1 is depressed, the voltage developed across a 2.49 megohm ± 1% resistor, connected between Pin H (+) and A (Gnd) of P5, shall be 500 ± 30VDC. The lamp shall flash when the pulse defined in Figure 11 is applied to P4.

3.6.2 Test Adapter Assembly 10555821 lighted.

a. Indicator DS1 shall be lighted when a voltage of 5.5 ± 0.3 volts is applied to Pin N of Plug P3, a voltage of 3.5 ± 0.3 volts is applied to Pin D of Plug P1, and Pin 1 of plug P2 is grounded. Indicator DS1 shall be extinguished when any of the following is placed at ground potential:

Pin e of Plug P4
Pin E of Plug P4
Pin G of Plug P2

b. Continuity shall exist between Pins O and o of Plug P4 when Switch S_1 is in position A.

c. Continuity shall exist between Pin o of Plug P4 and Pin M of Plug P2 when Switch S_1 is in position B.

d. Resistances specified below shall exist between the following pairs of pins.

RESISTANCE	BETWEEN
220K	Pin d of P4 and Pin I of P4
220K	Pin d of P4 and Pin K of P4
110K	Pin d of P4 and Pin F of P4
110K	Pin d of P4 and Pin C of P4
220K	Pin S of P4 and Pin H of P4
220K	Pin S of P4 and Pin D of P4
110K	Pin S of P4 and Pin J of P4
110K	Pin S of P4 and Pin L of P4
20K	Pin b of P2 and Pin Z of P2

e. Continuity.- The electrical continuity of the test adapter shall conform to Electrical Schematic D11728008.

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3.6.3 Cables.- Cables listed below shall meet the continuity requirements of their respective drawing as listed:

a.	A.C. Power Cable	10555804
b.	Battery Power Adapter Cable	11728018
c.	D.C. Power Adapter Cable	11728116
d.	QSTC Test Cable	10555801
e.	Coaxial Cable	10555802
f.	Coaxial Cable	10555803
g.	Cavity/PFN Test Adapter	11728017
h.	Coaxial Cable	11728019

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirements for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 test sets. The sample shall be manufactured in the same manner, using the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

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4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. Examination and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

(100% INSPECTION)

CLASS	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101.	Power ON	3.5.1	4.6.2.1
102.	Counter tester	3.5.2	4.6.2.2
103.	Stop Pre Amp and PM Tube tests	3.5.3	4.6.2.3
104.	Nixie power supply test	3.5.4	4.6.2.4
105.	Pulse forming network power supply test	3.5.5	4.6.2.5
106.	Photo multiplier power supply	3.5.6	4.6.2.6
107.	Q-Switch timing circuit	3.5.7	4.6.2.7
108.	Cavity	3.5.8	4.6.2.8
109.	Pulse forming network	3.5.9	4.6.2.9
110.	Trigger module	3.5.10	4.6.2.10
111.	High voltage divider	3.5.11	4.6.2.11
112.	PFN and Cavity Test Adapter 11728010	3.6.1	4.6.3.1
113.	Cavity tests	3.6.1.1	4.6.3.1.1
114.	PFN tests	3.6.1.2	4.6.3.1.2
115.	Test Adapter Assembly 10555821 lighted	3.6.2	4.6.3.2
116.	Cables	3.6.3	4.6.3.3

NOTE: The tests in Table I shall be conducted at a temperature between +70° and +80°F.

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4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- Three test sets shall be selected at random as a special sampling from each 50 produced. The samples shall meet the requirements and tests in Table II.

TABLE II

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Materials	3.1	Applicable Dwgs & Specs - Visual
302.	Fabrication	3.2	Applicable Dwgs & Specs - Visual
303.	General specifications	3.3	MIL-F-13926, MIL-P-46843, Visual

NOTE: The tests in Table II shall be conducted at a temperature between +70° and +80°F.

4.4.2 Environmental.- Three test sets shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall be subjected to the tests in Table III and shall then meet the requirements and tests in Table I.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
304.	Vibration	3.4.1	4.6.1.1
305.	Shock	3.4.2	4.6.1.2
306.	Storage temperature	3.4.3	4.6.1.3
307.	Immersion	3.4.4	4.6.1.4
308.	Operating temperatures	3.4.5	4.6.1.5

4.4.3 Failure of sample.- Should any item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

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4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- All inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the technical data package list.

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3 Test equipment.- In conjunction with 4.5.2.2 the following standard test equipment shall be utilized in the performance of the applicable tests specified in 4.6:

NOMENCLATURE	
1. Oscilloscope (CRO) with camera	Frequence response, D.C. to 85 megacycles; Rise time 4.5 nanoseconds; Input impedance 10,000 megohms or greater; Sensitivity 5 mv/cm; Sweep time 1 microsecond per centimeter, no parallax graticule; Camera lens f/1.9, 1.05 object to image ratio; Film, ASA equivalent of 10,000 (Polaroid Type 410 or equivalent.)

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	NOMENCLATURE	DESCRIPTION
2.	A.C. Power supplies	110 V.A.C. 60 Hz, 400 Hz ± 3%
3.	D. C. Power supplies	0 to 30 volts, 5 Amps ± 2% 0 to 8 volts, 1 Amp ± 2% 0-20,000 volts ± 10% 0-200 volts ± 2%
4.	Ammeter	0-250 millamps ± 2%
5.	Voltmeter	0-20,000 V.D.C. ± 5%, static type
6.	Digital voltmeter	Voltage measurements from ± 100 mv full scale to ± 1000V with the accuracy to be 1% of reading + (plus) one digit. Input resistance to be at least 10 megohms and constant on all ranges. The readout shall have in line digital display tubes and a polarity indication. The digital display shall have a storage feature which prevents the display from changing until a new measurement has been taken. The instrument shall have manual range selection with automatic decimal point positioning.
7.	Signal, square wave, generator	The signal generator shall produce a square wave burst of one cycle from an external sync. trigger signal from one cycle/per second to 5K cycles per second. The square wave signal amplitude shall be variable from one to 10 V.D.C. ± 1%. The rise time shall be less than 10 nanoseconds with a maximum of 5% overshoot and 1% symmetry error. The square wave duration shall be variable from 50 nano-seconds to 200 nanoseconds. The output impedance shall be less than 600 ohms.
8.	Electronic counter/preset unit	The counter shall be capable of measuring the time between N pulses of 0-10VPP and at least 50 nano-seconds duration. Where the first event opens the gate and the Nth

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NOMENCLATURE

	closes it, the input impedance shall be greater than 1 megohm, 50 pF shunt.
9. Universal bridge	Universal bridge shall be capable of measuring resistance, capacitance with in-line digital readout with automatic digit positioning. The range of capacitance measurement shall be 1 pF to 100 uF to 1% + 1 digit. The range of resistance shall be 10 ohms to 10 megohms to 1% plus one digit. The instrument shall not be damaged when zero resistance is being measured.
10. Calibrated photo-multiplier tube	The photo-multiplier tube shall be capable of measuring the output pulses of the light emitting diode specified in 3.5.3d and be designed to insert into the P.M. Tube test well. The output signal lead shall be adapted to connect to an oscilloscope to view the output signals. The calibration shall be traceable back to the N.B.S. standard.
11. High voltage pulse network	The high voltage pulse network shall be capable of producing the pulse shown in Figure 11 and be adopted to use during the test in 4.6.3.1.2b.

4.6 Test methods and procedures.

4.6.1 Environmental.- The test set shall be thoroughly examined for physical defects after each environmental test. A visual and tactile examination shall be made for missing parts and physical damage. Upon completion of each environmental tests, the test set shall be subjected to performance tests specified in 4.6.2 through 4.6.3.

4.6.1.1 Vibration.- The test set shall be positioned as specified in 3.4.4 and firmly mounted to the vibration tester. The test set shall be subjected to Test Method 514, Procedure I, Part I of MIL-STD-810, except that the acceleration, amplitude, frequency, and time duration shall be as specified in 3.4.1.

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4.6.1.2 Shock.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test shall be performed as specified in Procedure II of MIL-F-13926 except the shock pulse shall comply with the waveform, time duration and peak "G" value specified in 3.4.2.

4.6.1.3 Storage temperatures.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test shall be performed as specified in Procedure II of MIL-F-13926 at the temperatures specified in 3.4.3.

4.6.1.4 Immersion (leakage).- The test shall be performed in accordance with the immersion test defined in Procedure I of MIL-F-13926, except that immersion shall be maintained at the conditions specified in 3.4.4 for a minimum period of 12 minutes. When observing the test set during this time period, there shall be no evidence of bubbles from the test set.

4.6.1.5 Operational temperatures.- The test shall be performed in accordance with Procedure II of MIL-F-13926 at the extreme temperatures specified in 3.4.5 utilizing the test equipment indicated in 4.6.1.3. The test set shall be exposed to each specified temperature for a minimum period of 2 hours. At the end of each 2 hour period, and while still at the specified temperature, the test set shall be tested in accordance with 4.6.2 through 4.6.3.

4.6.2 Performance tests.- Each of the following performance tests shall be performed while the test set is supplied with power from a 28VDC source as specified in 4.6.2.1, and then repeated using a 110VAC, 60 Hz power source as specified in 4.6.2.1. All voltage measurements shall be made using a digital voltmeter and only when the "PUSH TO TEST" button is depressed, the power switch is ON, and "POWER ON" light is lit. All other switches should be turned to their open position and all adjustment knobs turned to their full counterclockwise position unless otherwise specified. The measurement of pulse characteristics shall be made using an oscilloscope with the necessary attachments to photograph the oscilloscope screen and be capable of the settings specified for horizontal time base, vertical sensitivity and triggering. The test equipment listed under 4.5.3 shall be used to perform all tests.

4.6.2.1 Power "ON".- Remove Cable Assembly 10555804 from test set storage compartment and connect it between the test set power receptacle and a 110VAC, 60 Hz power supply. When the power switch is turned on the "POWER ON" Indicator Lamp shall light. Depress the "PUSH TO TEST" button and indicator lamps shall meet the requirements of 3.5.1. Repeat the above procedure while supplying 110VAC, 400 Hz to the test set utilizing the same Cable 10555804. Remove Cable Assemblies 11728116 and 11728018 from test set storage compartment and connect them together. Connect these assembled cables between the test set power receptacle and a 28VDC power supply with the red lead on the (+) positive terminal and

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the black lead connected to the negative terminal. When the "POWER ON" switch is turned on and the "PUSH TO TEST" button pushed down the indicator lights shall function as specified in 3.5.1.

4.6.2.2 Counter tester.

4.6.2.2.1 Counter assembly receptacle.- The test set shall be energized as specified in 4.6.2.1. The test function switch (S2) shall be set to the "COUNTER" position and the "PUSH TO TEST" button depressed to its first position. Make the following measurements at the counter assembly receptacle:

a. Measure voltage at Pin F.

b. Connect a 15 ohms resistor between Pins F and B and read the panel meter, and remove resistor.

c. Measure voltage at Pin E.

d. Repeat the above readings when the "PUSH TO TEST" button is depressed to its second position. The results shall meet the requirements specified in 3.5.2.1a through 3.5.2.1b.

e. When the "READ" switch is set to the ON position and the High Voltage Interlock S5 closed, measure the voltage at Pin H and Pin N when the "PUSH TO TEST" button is depressed to its second position.

f. Using a 3VDC power supply connect the negative terminal to the chassis ground of the test set. Momentary apply 3VDC to Pin R and Pin x. The indicator light shall function as specified in 3.5.2.1. Place Pin W to chassis ground, indicator lamp shall light as specified in 3.5.2.1e.

g. With the power turned off the test set and the test function switch turned to the "COUNTER" position the following continuity checks shall be performed. Pins B, F, J and h connected to chassis ground and Pins C, D, S, T and L connected to the test points as specified in 3.5.2.1h thru 3.5.2.1m.

h. Turn the power on the test set and position the "TEST FUNCTION SWITCH" to the "COUNTER" position. Place the "READ SWITCH" to its "ON" position. The Range Switches S1 to S6 shall be in their "OFF" position. Position the range adjustment potentiometers to their full counterclockwise position and the range "STOP" and "START" potentiometers to their full clockwise position. Place the oscilloscope probe to Pin z of the counter assembly receptacle and adjust the oscilloscope to view and photograph the positive pulse as shown in Figure 1a. Connect the oscilloscope ground terminal to the chassis ground of the test set. Depress the "PUSH TO TEST" button down to its second and record the pulse from Pin z. The wave form shall be as specified in 3.5.2.1d and Figure 1a. Repeat the above procedure with the oscilloscope adjusted to measure the negative pulse shown in Figure 1a. Using a time interval counter set-up to measure the period between the positive pulse from Pin z and the negative pulse from Pin z, depress the "PUSH TO TEST" button to its second position. The time period shall be as specified in 3.5.2.1d and Figure 1a. Place the oscilloscope probe to Pin K of the counter assembly receptacle and adjust the oscilloscope to view and photograph the negative pulse as shown in Figure 1d. Connect the oscilloscope ground terminal to the test

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set chassis ground. Depress the "PUSH TO TEST" button to its second position and record the pulse from Pin K. The wave form shall be as specified in 3.5.2.1d and Figure 1d. Repeat the above procedure with the oscilloscope adjusted to measure the positive pulse as shown in Figure 1d. Using a time interval counter, set up to measure the period between the negative pulse from Pin K and the positive pulse from K depress the "PUSH TO TEST" button to its second position, the time period shall be as specified in 3.5.2.1d and Figure 1d.

With the oscilloscope probe placed on Pin A obtain a photograph of the screen. With the oscilloscope probe placed on Pin E obtain a picture of the oscilloscope screen when the "PUSH TO TEST" button is depressed to its second position, these pulses are shown on Figures 1b and 1c. The pulses shall meet the requirements specified in 3.5.2.1d. Repeat the above procedure when the "START" and "STOP" potentiometers are adjusted counterclockwise to obtain three amplitude readings of the "START" and "STOP" pulses from zero amplitude to maximum amplitude.

Connect a digital time interval counter gated to start on the start pulse from Pin "A" of the test set and stop on the stop pulse from Pin E to the test set. The range adjustment "STOP" and "START" potentiometers shall be turned to their full clockwise positions to obtain the maximum amplitude of the "STOP" and "START" pulses.

When the "PUSH TO TEST" button is depressed to its second position, measure the time interval between pulses from Pin "A" and Pin "E" for the following switch positions:

Turn "ON" the Range Adjustment Switch S1 and place its adjustment knob to its full counterclockwise position. All other range adjustment switches shall be positioned to their OFF position. Record the time between pulses at Pin A and Pin E. Repeat the above while the range adjustment knob is positioned for minimum, midrange and maximum adjustment. Repeat the above for each Range Adjustment Switch S1 through S6. The time intervals shall be as specified in 3.5.2.1d. When all range switches are turned on and the "PRESS TO TEST" button depressed to its second position, the number of pulses from Pin E shall be as specified in 3.5.2.1d. The above shall be repeated for any number of range switches turned to their "ON" position and the number of pulses from Pin E shall equal the number of range switches turned to their "ON" position.

4.6.2.2.2 Connector 10547210.- This test shall be performed with the test set energized as specified in 4.6.2.1, the test function switch turned to the "COUNTER" position. The "PUSH TO TEST" button depressed to its second position and held down. The following voltage measurements shall be taken.

a. Voltage at Pin C when Switch IX is set to Position A and Position B.
b. Voltage at Pin W.
c. Voltage at Pin Y and Pin D when Switch II is depressed and in its up position.

All the above voltages shall be as specified in 3.5.2.2a thru 3.5.2.2e.

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d. Connect a 1K Resistor from Pin O to chassis ground. Depress Switch II and hold. Connect an oscilloscope probe across the 1K Resistor and set trigger circuit on oscilloscope to (+) plus. Release Switch II and photograph the positive pulse on the oscilloscope screen. The pulse shall be as specified in 3.5.2.2f and as shown in Figure 2a.

e. Measure the voltage at Pin N when Switch VIII is set to Position A. Set Switch VIII in position "B" and measure the resistance between Pin N and chassis ground. The above readings shall be as specified in 3.5.2.2g.

f. Set Switch VI to its "AUTO" position and measure voltage at Pin F. Set Switch VI to "MAN" position. Connect the oscilloscope probe to Pin F and set trigger circuit of the oscilloscope to minus (-). Connect oscilloscope ground to the chassis ground of the test set. Depress Switch III and photograph the pulse on the oscilloscope screen. The pulse shall be as specified in 3.5.2.2g and Figure 2b.

g. Set the "READ" switch to its "ON" position and connect each Pin Q, z, o, E, M, C, A, N, B and P to chassis ground, one at a time. The digital indicator shall respond as specified in 3.5.2.2h.

h. This test shall be performed while the test set is not energized and the test function switch to its "COUNTER" position and the "PRESS" to test switch, depressed and held. The resistance from Pins H, J, L, and 1 to chassis ground shall be measured using a resistance bridge and shall be as specified in 3.5.2.21 thru 3.5.2.21

4.6.2.2.3 Connector 10547220.- This test shall be performed with the test set energized as specified in 4.6.2.1. The test function switch turned to the counter position. The "PUSH TO TEST" button depressed to its second position and held down. The following voltage measurements shall be taken at the pins indicated:

a. Pin C

Pin Q	When Switch II depressed
Pin Q	When Switch II released
Pin W	When Switch IV in position "A"
Pin u	When Switch IV in position "B"
Pin u & w	When Switch IV in its center position
Pin N	When Switch VIII in position "A"

The above voltage readings shall be measured by a digital voltmeter and be as specified in 3.5.2.3a thru 3.5.2.3g.

b. Position Switch VIII to "B" and using a resistance bridge or equivalent measure the resistance between Pin N and chassis ground. The resistance shall be as specified in 3.5.2.3g.

c. Place the oscilloscope probe on Pin K. Connect the oscilloscope ground to the test set chassis ground. Depress Switch I. Photograph the pulse on the oscilloscope screen. Hold Switch I down and measure the voltage at Pin Z and Pin G. The voltage shall be as specified in 3.5.2.3h and Figure 3a.

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d. Connect oscilloscope probe to Pin H and depress Switch II. Measure the voltage at Pin H. Release Switch II and photograph the resulting pulse as shown on oscilloscope screen. The voltage shall be as specified in 3.5.2.3i and shown on Figure 3b.

e. Place Switch VI in the manual (MAN) position. Measure the voltage at Pin J to chassis ground. Place oscilloscope probe at Pin J. Depress Switch III and photograph the resulting pulse on the oscilloscope screen. The voltage wave form shall be as shown in Figure 3c. Release Switch III and place Switch VI in the "AUTO" position. The train of pulses as viewed on the oscilloscope screen shall be as specified in 3.5.2.3j and Figure 3d.

f. Connect each of the following to chassis ground N, Y, o, X, n, V, K, i and S. View the digital indicator when the "READ" switch is in its ON position. The digital indicator shall function as specified in 3.5.2.3l.

g. Measure the resistance between the following pins and chassis ground, using a resistance bridge and the "PUSH TO TEST" button up. Pins A, L, B, D, o, f, d, F and 1. The resistance shall be as specified in 3.5.2.3m.

4.6.2.2.4 Connector 10547230.- This test shall be performed with the test set energized as specified in 4.6.2.1. The test function switch turned to the "COUNTER" position, the "PUSH TO TEST" button depressed to its second position and held down. The following voltage measurements shall be taken at the pins indicated:

a. Pin O When Switch I depressed and released
Pin o
Pin K When Switch VIII is in position "A"

The above voltage measurements shall be as specified in 3.5.2.4a, b, e

b. Place the oscilloscope probe at Pin J. Set Switch IV to position "A". The resulting pulse from Pin J shall be photographed and be as specified in 3.5.2.4c and Figure 4a.

c. Place theoscilloscope probe at Pin I. Set Switch IV to position "B". The resulting pulse from Pin I shall be photographed and be as specified in 3.5.2.4d and Figure 4b.

d. Place Switch VIII to position "A". Measure the voltage at Pin K, place Switch VIII to position "B". The resistance from Pin K to chassis ground shall be measured by a resistance bridge. The voltage and resistance readings shall be as specified in 3.5.2.4e, 3.5.2.4f.

e. Place the oscilloscope probe to Pin E, connect the oscilloscope ground to the test set chassis ground, set Switch VI to the "AUTO" position. The pulses shown on the oscilloscope screen shall be as specified in 3.5.2.4g and Figure 4d.

f. Place the oscilloscope probe to Pin E. Connect the oscilloscope ground to the test set chassis ground. Set Switch VI to "MAN" position. Depress Switch III and photograph the resulting pulse on the oscilloscope screen. Repeat the above three times. The pulse wave form shall be as specified in 3.5.2.4h and Figure 4c.

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g. Place the READ switch in the "ON" position. With a digital voltmeter measure the voltage at Pin u. Alternately connect each Pin b, a, S, w, R, V, Q, Z, P and T to chassis ground. The digital indicator shall function as specified in 3.5.2.4j.

h. Turn power off the test set. Using a resistance bridge. Measure the resistance between Pins G, Y, X, N, B, L, D, F, C and K and chassis ground. The resistances shall be as specified in 3.5.2.4k.

4.6.2.2.5 Connector 10547240.- This test shall be performed with the test set energized as specified in 4.6.2.1, the test function switch turned to the "COUNTER" position. The "PUSH TO TEST" button depressed to its second position and held down. The following voltages shall be measured at the pins indicated and switches in position specified:

a. Pin K
Pin G
Pin A
Pin K When Switch II depressed and released
Pin O When Switch VIII is set to position "A"
Pin X When Switch VI is set to "MAN" position

The above voltage measurements shall be as specified in 3.5.2.5a thru 3.5.2.5c, 3.5.2.5e and 3.5.2.5o.

b. Place the oscilloscope probe on Pin B. Connect oscilloscope ground to the test set chassis ground. Depress Switch II and release. Photograph the pulse on the oscilloscope screen. The voltage wave form shall be as specified in 3.5.2.5f and Figure 5a.

c. Place the oscilloscope probe on Pin b. Connect oscilloscope ground to the test set chassis ground. Depress Switch II and release. Photograph the resulting wave form on the oscilloscope screen and it shall be as specified in 3.5.2.5g and Figure 5b.

d. Place the oscilloscope probe on Pin V. Connect the oscilloscope ground to the test set chassis ground. Place Switch IV to position "B". Photograph the resulting wave form on the oscilloscope screen and it shall be as specified in 3.5.2.5h and Figure 5c.

e. Place the oscilloscope probe on Pin R. Connect the oscilloscope ground to the test set chassis ground. Place Switch IV from position "B" to position "A". The resulting wave form shall be photographed and it be as specified in 3.5.2.5i and Figure 5d.

f. Position Switch V to "P". Measure the voltage at Pins P, Q, W and Y with a digital voltmeter. Repeat the above with Switch V set to positions Q, W and Y. The voltage shall be as specified in 3.5.2.5j.

g. Set Switch VII to position "A". Measure the resistance from Pin E to chassis ground. Repeat when Switch VII is in position "B". The readings shall be as specified in 3.5.2.5k and 3.5.2.5l. With Switch VIII in position "B" measure the resistance from Pin O to chassis ground. The resistance shall be as specified in 3.5.2.5m and 3.5.2.5n.

h. Place the oscilloscope probe on Pin c. Connect oscilloscope ground to the test set chassis ground. Depress Switch III and release. Photograph the resulting negative pulse on the oscilloscope screen. The voltage wave form shall be as specified in 3.5.2.5p and Figure 5e.

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i. Apply 2.0 ± 0.5VDC to Pin L. Measure the voltage from Pin K using a digital voltmeter. Disconnect the applied voltage from Pin L and again measure the voltage from Pin K. The voltages measured shall be as specified in 3.5.2.5r.

j. Place the "READ" switch to the "ON" position. Alternately connect Pins Z, a, h and V to chassis ground. The digital indicator shall function as specified in 3.5.2.5s.

k. Alternately place Pins I, H, F, J and d respectively to chassis ground. The Neon Indicator Lamps shall function as specified in 3.5.2.5.

l. Measure the resistance from Pin M to chassis ground. Repeat for Pin e and Pin o. The resistance shall be as specified in 3.5.2.5t thru 3.5.2.5w.

m. Continuity shall be measured using a standard volt ohmmeter for the following pins and their respective test point:

Pin C	T.P. #17
Pin e	T.P. #21
Pin J	T.P. #22
Pin T	T.P. #23
Pin N	T.P. #24
Pin D	T.P. #27

The above readings shall be as specified in 3.5.2.5x.

n. Depress the lever switch. The indicator lamps shall function as specified in 3.5.2.5y.

4.6.2.3 Stop Pre-amp and P.M. Tube tests.- This test shall be performed with power supplied to the test set as specified in 4.6.2.1, with the test function switch set on the "STOP PRE-AMP AND P.M. TUBE" position. All measurements shall be made from output receptacle J10 and output Jack J14 (Pre-amp Output).

a. With the "PUSH TO TEST" button depressed to its second position and held down, measure the voltage at Pin "A". Release the "PUSH TO TEST" button. Connect a variable resistance of 50-100 ohms ± 10% to Pin "A" and chassis ground. Connect an ammeter to measure the current from Pin A through the resistance load. Depress the "PUSH TO TEST" button to its second position. Vary the load resistance and record the current from Pin "A". It shall be as specified in 3.5.3a.

b. Connect a 180 ohm ± 5% load resistance from Pin "C" to chassis ground. Place oscilloscope probe across this resistance. Set oscilloscope and adjustments to measure the voltage wave form shown in Figure 6a. Depress the "PUSH TO TEST" button down and release. Photograph the pulse as seen on the oscilloscope screen. It shall be as specified in 3.5.3b and Figure 6a.

c. Connect the oscilloscope input lead to Jack J14 (Pre-amp Input). Turn the separation and amplitdue knobs to their full clockwise position. Set the oscilloscope adjustments to measure the pulses shown in Figure 6b. Depress the push to test button to its second position and

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hold down. Adjust the oscilloscope to achieve the best trace on the screen. Record the frequency of the pulse pattern, the amplitude of the pulses, and the pulse duration. Vary the amplitude knob counterclockwise and record the change in pulse amplitude. Vary the separation knob counterclockwise and measure the change in separation of the pulses. Repeat the above procedure for the amplitude knob in its minimum midrange and maximum position. The above measurements shall be as specified in 3.5.3c and Figure 6b.

d. This test shall be performed by using a calibrated photo-miltiplyer tube. The tube shall be capable of measuring the light pulse as specified in 3.5.3d. The output of the calibrated photomultiplyer tube shall be connected to an oscilloscope isolated from the test set chassis ground. With the power supplied as specified, depress the "PUSH TO TEST" button to its second position and hold down. Insert the calibrated P.M. tube into the test well. Adjust the oscilloscope for the best trace on the oscilloscope screen. Record the frequency of the light pulse, the amplitude, and duration of the pulses from the calibrated photomultiplyer tube. Vary the separation knob fully clockwise and full counterclockwise and record pulse separation as measured by the oscilloscope trace. Vary the amplitude knob fully clockwise and fully counterclockwise and note any change in pulse amplitude. The measured values shall be as specified in 3.5.3d.

e. Connect a high voltage cable fitted to plug into the high voltage output jack to a high voltage static voltmeter capable of measuring 0-20,000VDC ± 10% negative potential. Depress the interlocking switch in the P.M. Tube Receptacle and record the voltage reading. Remove the H. V. interlock and measure the voltage output. Repeat the above procedure three times. The high voltage shall be as specified in 3.5.3e.

f. With the power turned off the test set, measure the resistance from Pin E to chassis ground and the resistance between Pin E of Connector J10 to the center contact of the "PRE-AMP OUTPUT" connector. The resistance shall be as specified in 3.5.3f.

g. Using a resistance bridge, measure the resistance between "PM TUBE OUTPUT" connecdor to chassis ground, and from the center contact of the "PM TUBE OUTPUT" connector to the center contact of the "OUTPUT" connector, the measured values shall be as specified in 3.5.3g.

4.6.2.4 Nixie power supply test.- This test shall be performed when the test set is energized as specified in 4.6.2.1. With the test function switch in "NIXIE POWER SUPPLY" position.

a. With a digital voltmeter measure the voltage from Pin A and Pin X to Pin B. With a resistance bridge measure the resistance between Pin B and Pin D of receptacle J4 and the continuity between Pin B and Pin D to chassis ground. The measured values shall be as specified in 3.5.4a and 3.5.4d.

b. With the "PUSH TO TEST" button in its open position, turn the test function switch to the counter position, to remove the test panel meter from circuit. Using a resistance bridge measure the resistance from Pin E of J4 to chassis ground. The reading shall be as specified in 3.5.4c.

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c. Set the test function switch to the "NIXIE POWER SUPPLY" position. Connect a 160VDC power supply between Pin A of J4 and chassis ground. With the plus (+) terminal to Pin "A" do not depress the "PUSH TO TEST" button. Set voltage to the test set from the power supply to 160VDC ± 1VDC and record the panel meter deflection. The measured value shall be as specified in 3.5.4.6b.

4.6.2.5 Pulse forming network power supply.

a. The power shall be supplied as specified in 4.6.2.1 and the test function switch set to the "P.F.N. POWER SUPPLY" position. Using a digital voltmeter measure the voltage at Pin H of the J7 Receptacle. The reading shall be as specified in 3.5.5a. Depress the "PUSH TO TEST" button to its first position and hold. Using a digital voltmeter measure the voltage from Pin A to Pin B and the voltage between Pin C and Pin F of J7 Receptacle. The readings shall be as specified in 3.5.5b. Depress the "PUSH TO TEST" button to its second position and hold down. Using an impedance meter, measure the impedance from the connector body of J5 to Pin D of the J7 Receptacle. Release the "PUSH TO TEST" button and measure the impedance from the connector body of J5 to the chassis ground of the test set. The impedance shall be as specified in 3.5.5d. Measure the resistance between Pin K of J7 and chassis ground. It shall be as specified in 3.5.5g.

b. Connect a 5.5 ± .3VDC power supply to Pin L of the J7 Receptacle. With the plus terminal to Pin L and the negative terminal to chassis ground, depress the "PUSH TO TEST" button down to its second position and note Indicator Lamp "A". The response shall be as specified in 3.5.5c.

c. Connect a 1500VDC ± 5% high voltage power supply with the positive terminal connected to center conductor of J5 and the negative terminal connected to the chassis ground of the test set. Depress the "PUSH TO TEST" button down to its second position and hold down. Increase the voltage from the power supply until the voltage as measured by a digital voltmeter connected across the power supply is 1500VDC ± 1%. Note the panel meter deflection. With the full potential applied to J5, switch the power supply switch off and measure the time that the deflection of the panel meter goes to zero. Release the "PUSH TO TEST" button. The above shall be as specified in 3.5.5e and 3.5.5f.

4.6.2.6 Photomultiplier power supply.- This test shall be performed while the test set is energized as specified in 4.6.2.1. The test function switch in the "P.M. POWER SUPPLY" position.

a. Depress the "PUSH TO TEST" button to its first position and hold. Using a digital voltmeter, measure the voltages at Pin "A" to Pin "B" of Receptacle J9 and at Pins c and E. The voltage shall be as specified in 3.5.6a.

b. Depress the "PUSH TO TEST" button to its second position and hold down, place an oscilloscope probe on Pin E of Receptacle J9. Connect the oscilloscope ground to chassis ground of the test set. Adjust the oscilloscope to view and photograph the pulse shown on Figure 7a.

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Release the "PUSH TO TEST" button to the first contact position and photograph the resulting wave form on the oscilloscope screen. Keeping the same adjustments of the oscilloscope, place the oscilloscope probe on Pin C of J9 Receptacle. Depress the "PUSH TO TEST" button to the second position and photograph the resulting wave form on the oscilloscope screen. The pulses shall be as specified in 3.5.6b and shown in Figures 7a and 7b.

c. Connect a 1500VDC power supply to the J8 Receptacle with the negative (-) terminal applied to the center conductor and the plus (+) side connected to the connector body. Depress the "PUSH TO TEST" button and hold down. Note the meter deflection. Release the "PUSH TO TEST" button. Increase the voltage from the power supply until a reading of 1500VDC ±1% is obtained as measured by a digital voltmeter across the power supply. Record the panel meter deflection. The meter response shall be as specified in 3.5.6c.

4.6.2.7 Q-Switch timing circuit.- This test shall be performed while the test set is energized as specified in 4.6.2.1. The test function is set on the Q.S.T.C. position.

a. Depress the "PUSH TO TEST" button to its first position and hold. Using a digital voltmeter measure the voltage between Pin E and Pin B of Receptacle J6. The voltage shall be as specified in 3.5.7a. Place an oscilloscope probe on Pin A of J6. Adjust oscilloscope to view and photograph the pulse shown in Figure 8a. Depress the "PUSH TO TEST" button to its second position. Photograph the resulting wave form on the oscilloscope screen and shall it be as specified in 3.5.7b and shown in Figure 8a.

b. Connect a 20VDC power supply to Pin D and Pin C. With the positive (+) terminal at Pin D and the negative terminal to Pin C of J6 Receptacle. Place an oscilloscope probe to Pin J. Connect the oscilloscope ground to the bhassis ground of the test set. Adjust the oscilloscope to view the pulse shown on Figure 8b. Depress the "PUSH TO TEST" button and hold down. The pulses shown on the oscilloscope screen shall be as specified in 3.5.7c and shown in Figure 8b.

c. Connect a 3.0 ± 0.3VDC power supply momentarily to Pin F and the center conductor of J17. Momentarily ground Pin H. The indicator lights shall function as specified in 3.5.7d and 3.5.7e.

4.6.2.8 Cavity.- Set the test function switch to "CAVITY" position. Measure the voltage between Pin J and chassis ground, and between Pin F and chassis ground, using a digital voltmeter, The voltages shall be as specified in 3.5.8. Place Pin K of J18 Receptacle to chassis ground. Note meter deflection. Depress the "PUSH TO TEST" button to its first position and measure the voltage from Pin D to chassis ground. Depress "PUSH TO TEST" button to its second position and measure the voltage at Pin J of Receptacle J8. The above voltage readings shall be as specified in 3.5.8.

4.6.2.9 Pulse forming network.- Set the "TEST FUNCTION SWITCH" to pulse forming network. Position and energize the test set as specified in 4.6.2.1. Depress the "PUSH TO TEST" button to the first contact position and hold. Measure the voltage from Pin E of Receptacle J18 to chassis ground, using a digital voltmeter. The reading shall be as specified in 3.5.9a.

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a. Place an oscilloscope probe on Pin L of J18. Connect oscilloscope ground to the chassis ground of the test set. Adjust the oscilloscope to view and photograph a positive pulse as shown on Figure 9. Depress the "PUSH TO TEST" button to its second position and release. Photograph the wave form on the oscilloscope screen. The pulse shall be as specified in 3.5.9b and shown on Figure 9.

b. Connect a constant current 250 micro-ampere source to Pin H of J18. With the positive terminal to Pin H and the negative terminal to chassis ground, note the meter deflection. The meter response shall be as specified in 3.5.9c.

4.6.2.10 Trigger module.- The test set shall be energized as specified in 4.6.2.1. The test function switch set in the "TRIGGER" position and the high voltage interlock in its closed position. Depress the "PUSH TO TEST" button to its first position and hold. With a digital voltmeter measure the voltage across the contacts of terminal Block E1. Depress the "PUSH TO TEST" button to its second position and hold. Measure the voltage at Connector J3 and record the reading. It should be as specified in 3.5.10a. Place an oscilloscope probe on Connector J3. Connect the oscilloscope ground to the chassis ground of the test set. Adjust the oscilloscope to view and photograph a positive pulse as shown in Figure 10. Release the "PUSH TO TEST" button. The wave form as seen on the oscilloscope screen shall be as specified in 3.5.10b.

a. Connect a 1000 ± 10VDC power supply to Terminal P2. The neon Bulb E shall function as specified in 3.5.10c.

4.6.2.11 High voltage divider.- The test set shall be energized as specified in 4.6.2.1. The test function switch set to the "H.V. DIVIDER" position, and the high voltage interlock closed.

a. Depress the "PUSH TO TEST" button to the second position and hold. Use a digital voltmeter to measure the voltage from P3 to chassis ground. It shall be as specified in 3.5.11a.

b. Connect Terminal J16 to chassis ground. The Indicator Lamp D shall respond as specified in 3.5.11a.

c. Release the "PUSH TO TEST" button and connect a 160 ± 5VDC power supply across Terminal Block E2. Note the panel meter deflection. The reading shall be as specified in 3.5.11b.

4.6.3 Accessory equipment.- The following tests shall be performed without the use of Test Set 10555677.

4.6.3.1 P.F.N and Cavity Test Adapter 11728010.

4.6.3.1.1 Cavity test circuit.- Connect a 5VDC ± .1VDC power supply to Receptacle P5 with the positive terminal to Pin F and negative terminal to Pin A. Connect a 28VDC ± 1VDC power supply to P5 Receptacle. With the positive terminal to Pin D and the negative terminal to Pin A, using a digital voltmeter measure the voltage from Pin D to Pin E of J2. The measured voltage shall be as specified in 3.6.1.1b.

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a. Connect a square wave generator to Pin B of Receptacle J2 and Pin A of J2 (ground). The output of the square wave generator shall be a train of square waves with an amplitude of ± 3.5VDC ± .1VDC and 100 nano-second duration and spaced apart at least 2.5 milliseconds. The time between each square wave cycle shall be adjustable from 1.5 milliseconds to 5 milliseconds. An oscilloscope shall be used to view these square wave pulses. Vary the square wave input pulse from 5 milliseconds to 2 milliseconds and determine the time period between these square waves to have indicator lamp to light. The time measured shall be as specified in 3.6.1.1a.

4.6.3.1.2 P.F.N. test circuit.

a. To perform this test, connect a 28 ± 1VDC power supply to Receptacle P5. With the positive terminal to Pin M and the negative terminal to Pin A, connect a 75 ± 5% microfarad capacitor to the output of the potted Transformer 10540823 of the adapter under test. Connect a digital voltmeter across this capacitor to read the voltage build-up and decay. Turn voltage power supply on and adjust to 28VDC ± 1VDC. Push the S1 Switch and watch the voltage build-up while the 75 ± 5% capacitor is charging. Measure the maximum voltage on capacitor and the time to charge the capacitor. These measurements shall be as specified in 3.6.1.2a. While holding S1 down, the voltage across the capacitor shall not exceed 18000VDC. If this voltage exceeds the value specified, release S1 and terminate the test.

b. When the capacitor is fully charged as specified above and meets the requirements of 3.6.1.2, release Switch S1. Connect a 2.49 megohm ± 1% resistor between Pin H and Pin A of P5. Connect a digital voltmeter across this resistance. Depress S1 and read the voltage across the 2.49 megohm resistor. It shall be as specified in 3.6.1.2a. Remove the 28VDC power from adapter. Measure the resistance between the contacts of P3 when S1 is open and closed. The measured resistance shall be as specified in 3.6.12b. Connect the charged capacitor to the P3 Connector. With the plus terminal of the capacitor to Pin A, connect the special pulse circuit which can deliver the pulse shown in Figure 11 to Connector P4. Turn on the 28VDC power supply and push Switch S1. Apply the pulse shown in Figure 11 to P4. The response shall be as specified in 3.6.1.2c.

4.6.3.2 Test Adapter Assembly 10555821.- Connect a 5.5 ± .3VDC power supply to P3, with the positive terminal to Pin N and negative terminal to ground. Connect another power supply of 5.5 ± .3VDC to P1. With the positive terminal to Pin D, place Pin 1 of Plug P2 at ground potential. While keeping the above connections, ground the following pins and note the indicator light.

Pin e of Plug P4
Pin E of Plug P4
Pin G of Plug P2

The indicator light shall function as specified in 3.6.2a.

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a. Using a standard volt-ohm meter check the continuity between the following:

Pin O and Pin o of P4
Pin o of P4 and Pin M of Plug P2 when S1 is in position "B"

The above shall be as specified in 3.6.2b and 3.6.2c.

b. Using a resistance bridge and all power "OFF" the test adapter, measure the resistances specified in 3.6.2d.

c. Using a standard volt-ohm meter, check the continuity as specified in 3.6.2e.

4.6.3.3 Cables assemblies.- The cables listed in 3.6.3 shall be tested for continuity using a standard volt-ohm meter.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10555677. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The test set is intended for use as direct and general support maintenance for the certification of suspected defective rangefinder modules.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Selection of applicable levels of preservation, packaging and packing (see 5.1).
c. Applicable packaging data sheet number (see 5.1).
d. Provisions for first article testing.

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PIN Z

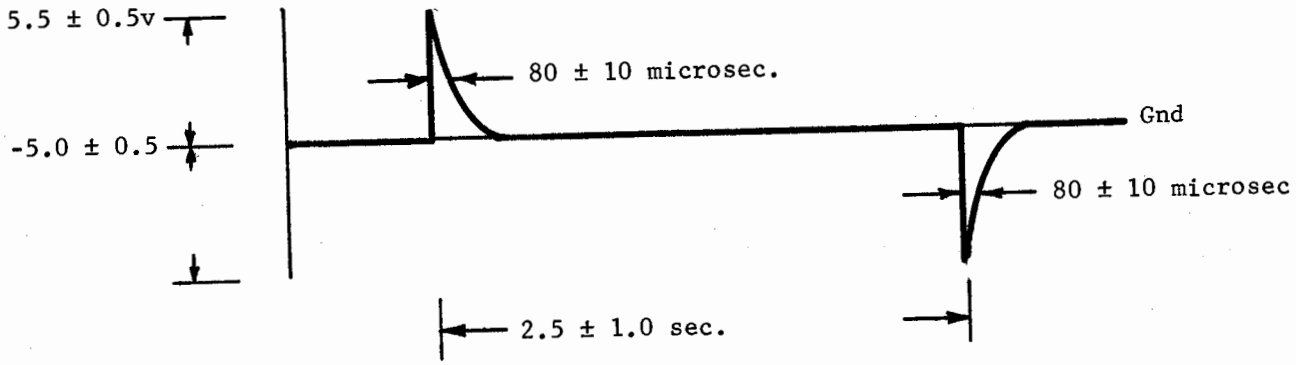


FIGURE 1a

PIN A

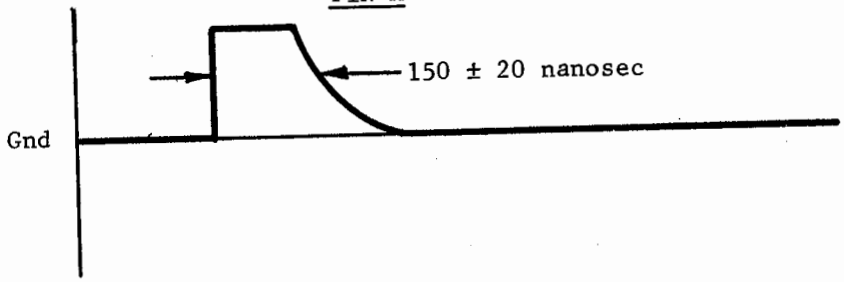


FIGURE 1b

PIN E

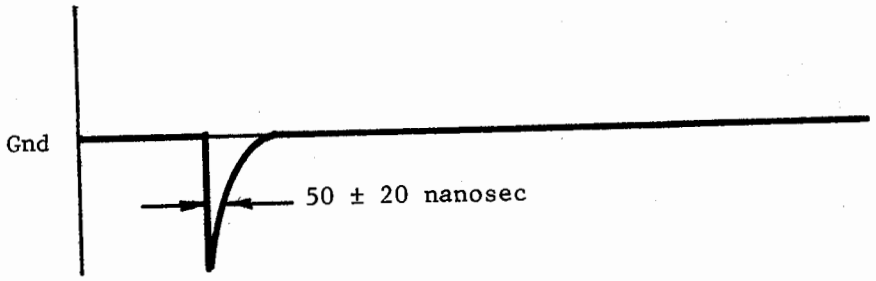


FIGURE 1c

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PIN K

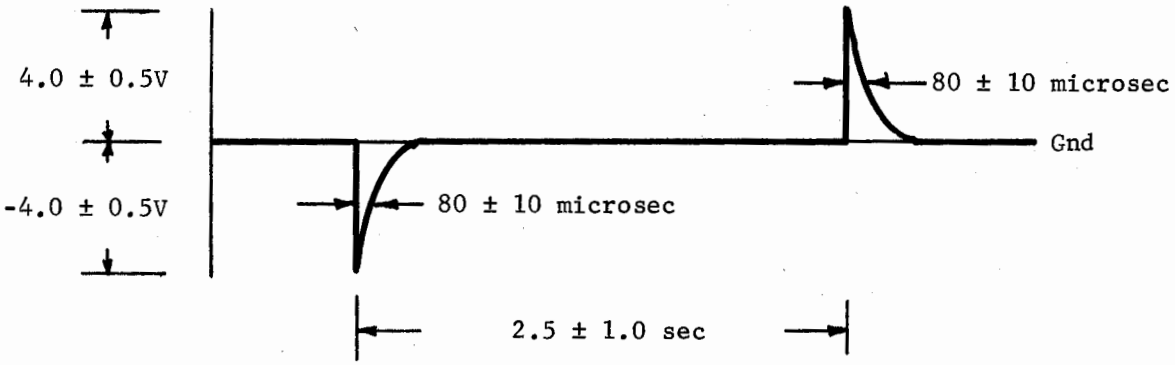


FIGURE 1d

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PIN O

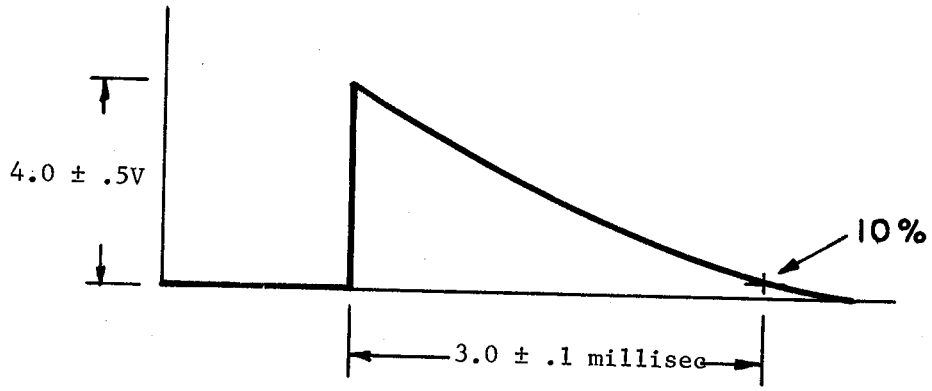


FIGURE 2a

PIN F

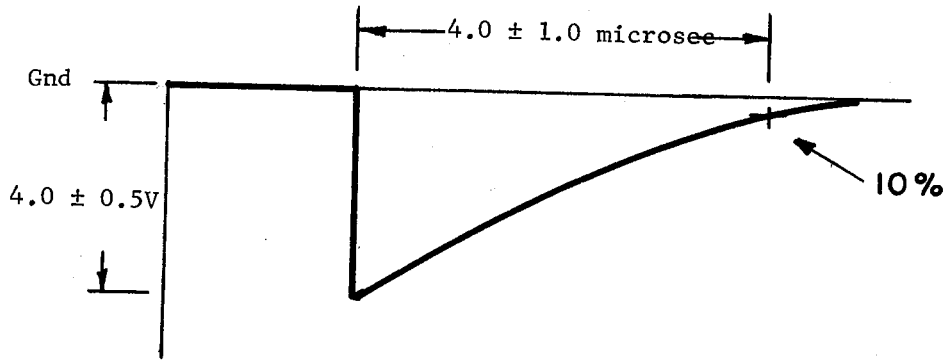


FIGURE 2b

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PIN K

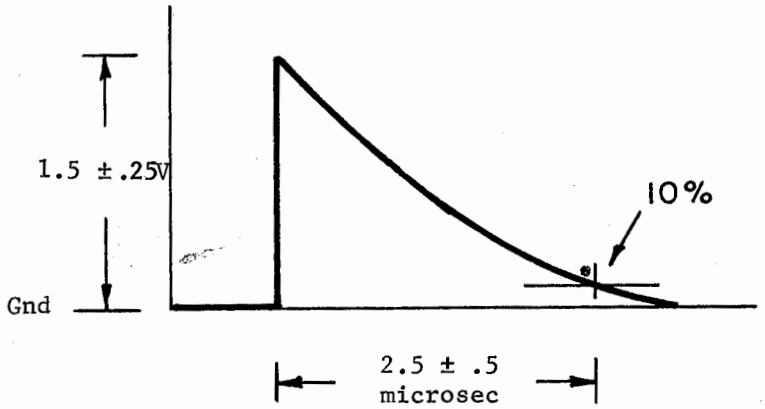


FIGURE 3a

PIN H

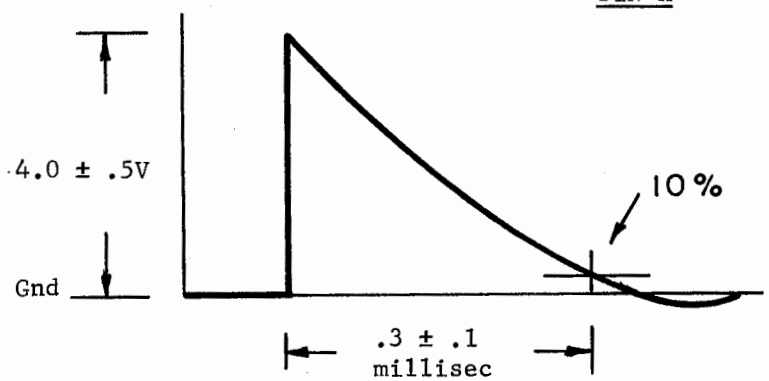


FIGURE 3b

PIN J

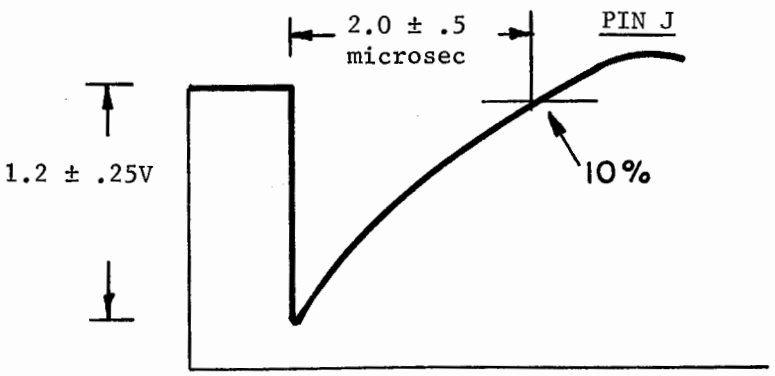


FIGURE 3c

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PIN J

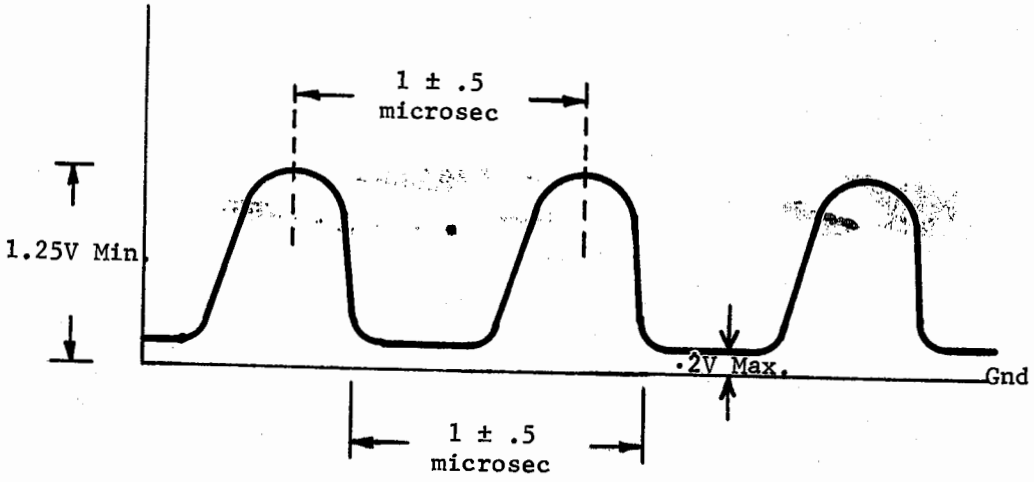


FIGURE 3d

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PIN J

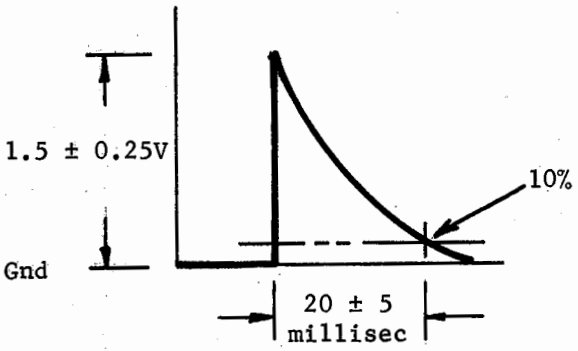


FIGURE 4a

PIN I

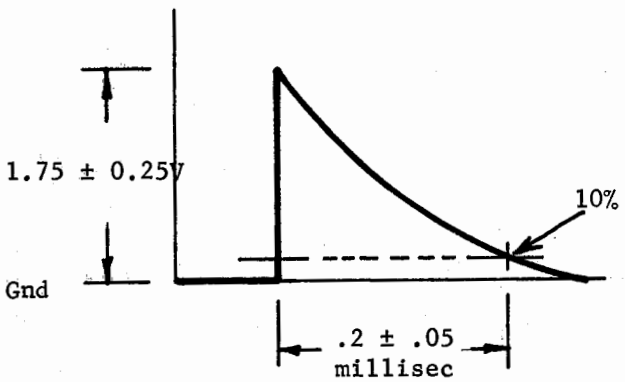


FIGURE 4b

PIN E

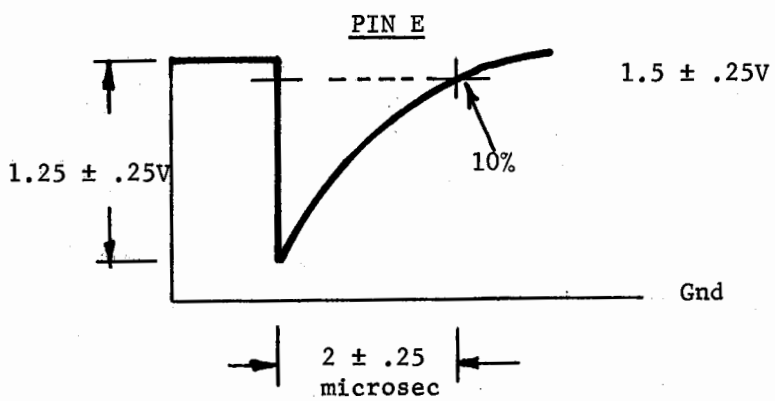


FIGURE 4c

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
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PIN E

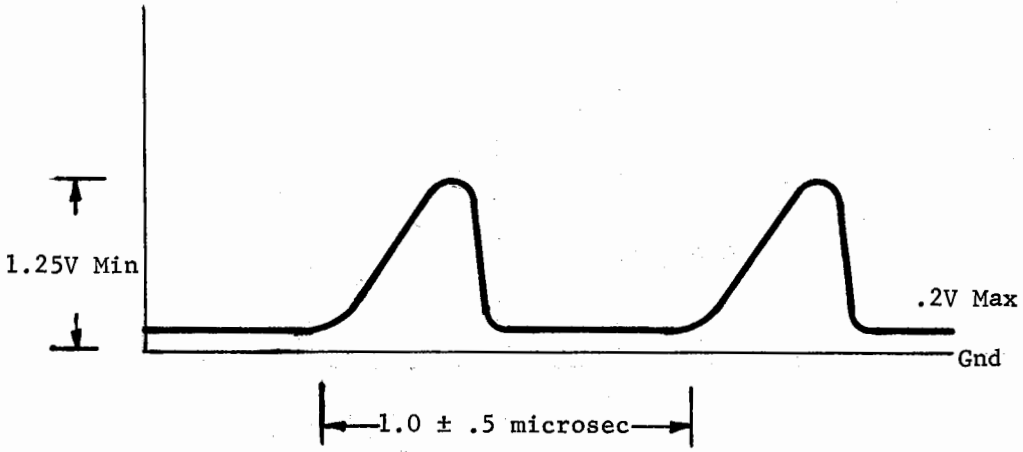


FIGURE 4d

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
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PIN B

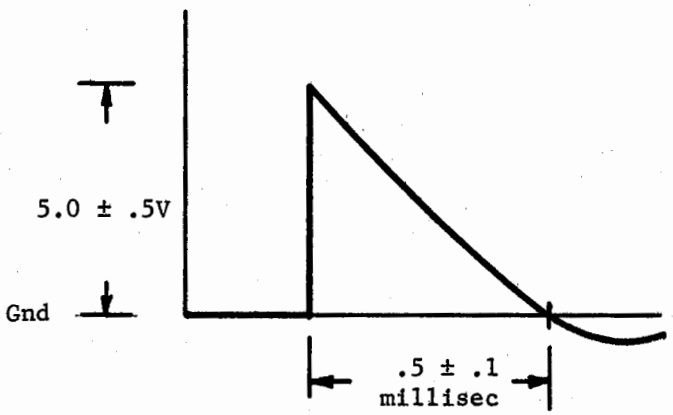


FIGURE 5a

PIN b

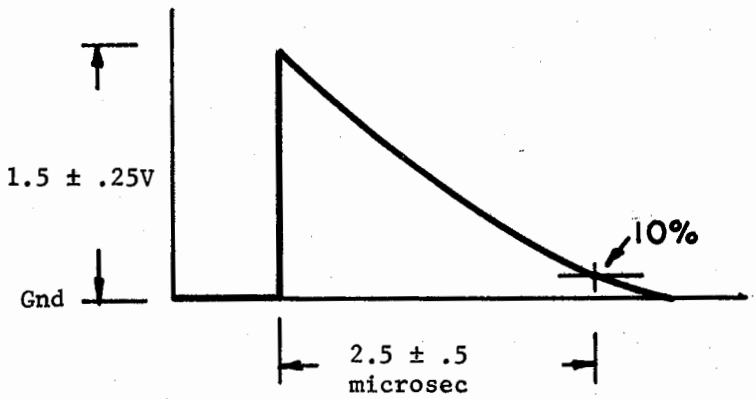


FIGURE 5b

PIN V

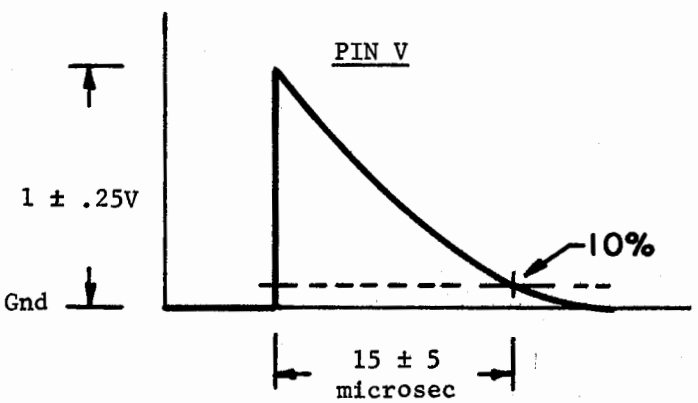


FIGURE 5c

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
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PIN R

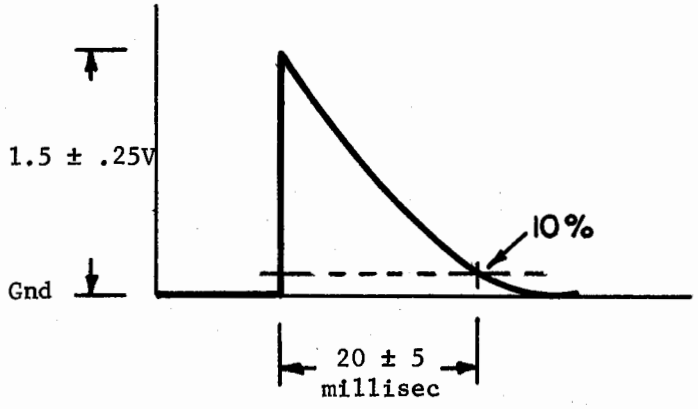


FIGURE 5d

PIN C

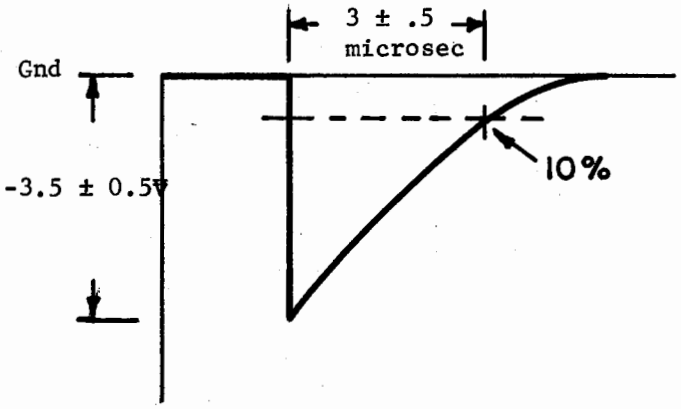


FIGURE 5e

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
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PIN C

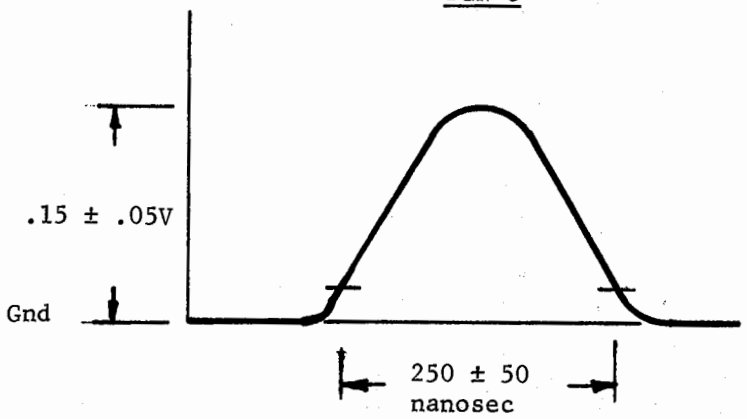


FIGURE 6a

PIN J14

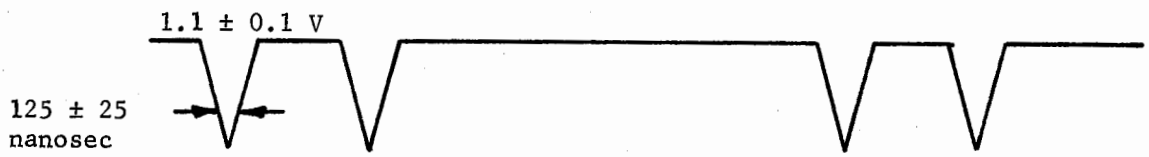


FIGURE 6b

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
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PIN E

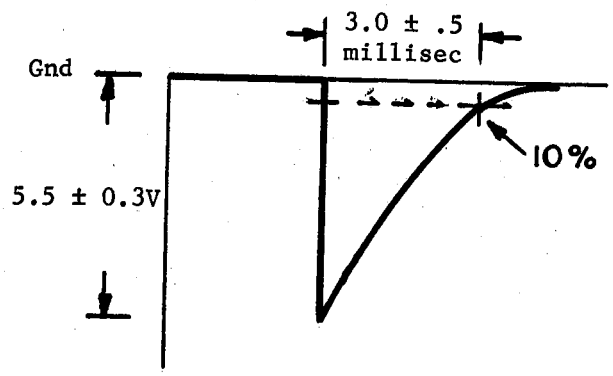


FIGURE 7a

PIN C

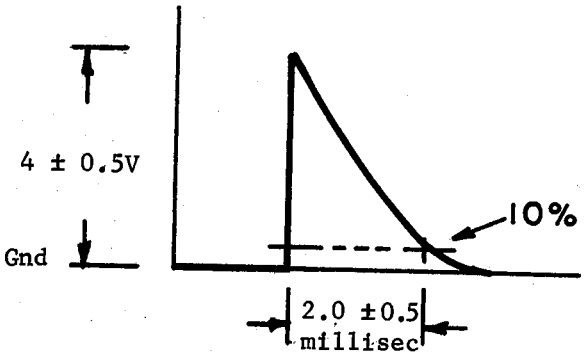


FIGURE 7b

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
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PIN A

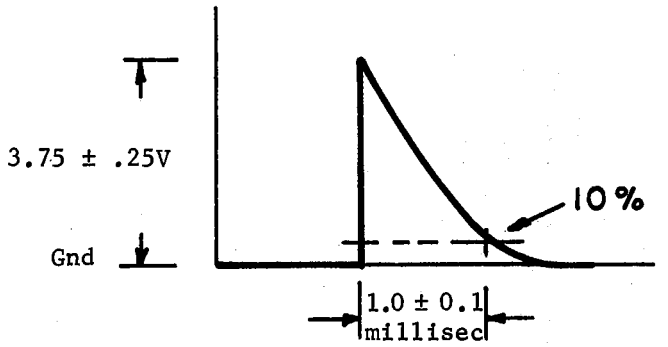


FIGURE 8a

PINS J & L

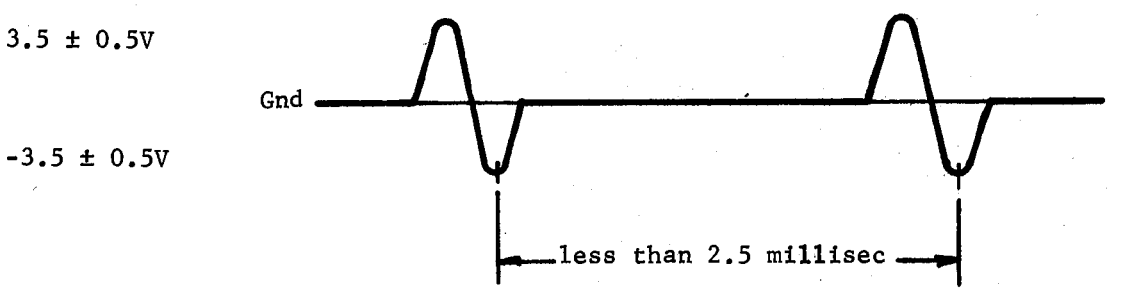


FIGURE 8b

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

PIN L

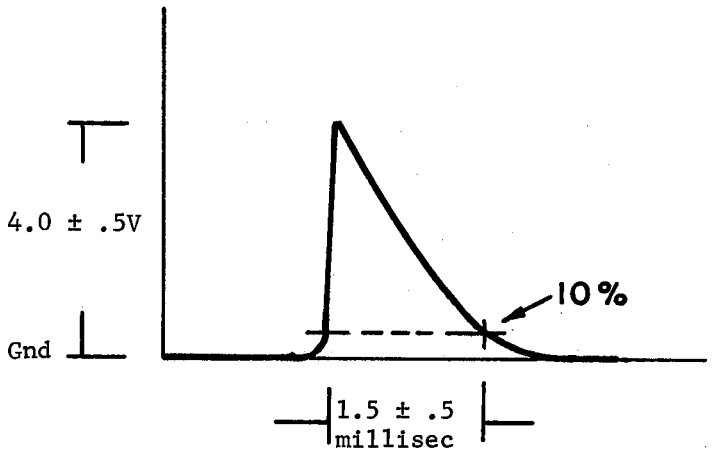


FIGURE 9

PIN J_3

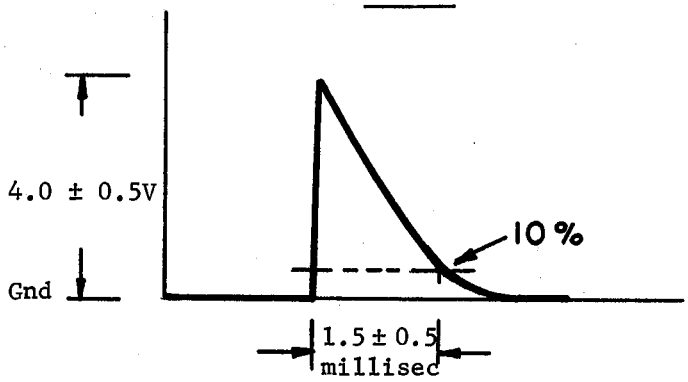


FIGURE 10

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
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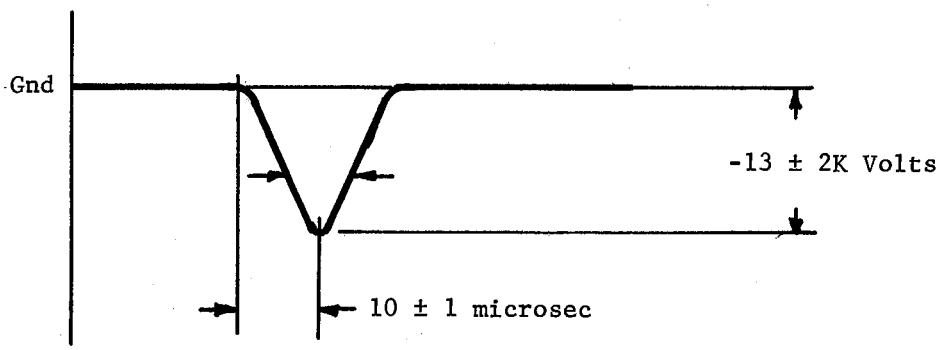


FIGURE 11

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